

|                                                                                                                                                                                                                                                                                 |      |     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|
| <p>1 (a) force proportional to product of the (two) masses and inversely proportional to the square of their separation<br/> <i>either</i> reference to point masses <i>or</i> separation &lt;&lt; 'size' of masses</p>                                                         | M1   |     |
| (b) gravitational force provides/is the centripetal force                                                                                                                                                                                                                       | B1   | [2] |
| $GMm/r^2 = mv^2/r$ or $GMm/r^2 = mr\omega^2$ and $v = r\omega$<br>and algebra leading to $v = (GM/r)^{1/2}$                                                                                                                                                                     | B1   | [2] |
| <p>(c) (i) 1. <math>v_A/v_B = (r_B/r_A)^{1/2}</math><br/> <math>= (2.2 \times 10^{10}/1.3 \times 10^8)^{1/2}</math><br/> <math>= 13</math> (13.0)</p>                                                                                                                           | C1   | [2] |
| <p>2. <math>v = 2\pi r/T</math> or <math>v \propto r/T</math> or <math>vT/r = \text{constant}</math></p>                                                                                                                                                                        | C1   |     |
| $T_A/T_B = (r_A/r_B) \times (v_B/v_A)$<br>$= (1.3 \times 10^8/2.2 \times 10^{10}) \times (1/13)$<br>$= 4.5$ (4.54) $\times 10^{-4}$                                                                                                                                             | C1   | [2] |
| or                                                                                                                                                                                                                                                                              |      |     |
| $T^2 = 4\pi^2 r^3/GM$ or $T^2 \propto r^3$ or $T^2/r^3 = \text{constant}$                                                                                                                                                                                                       | (C1) |     |
| $T_A/T_B = (r_A^3/r_B^3)^{1/2}$<br>$= [(1.3 \times 10^8)/(2.2 \times 10^{10})^3]^{1/2}$<br>$= 4.5$ (4.54) $\times 10^{-4}$                                                                                                                                                      | (C1) | [3] |
| <p>(ii) <math>T = 2\pi/1.7 \times 10^{-4}</math><br/> <math>= 3.70 \times 10^4 \text{ s}</math><br/> <math>T_B = 3.70 \times 10^4/4.54 \times 10^{-4}</math><br/> <math>= 8.1 \times 10^7 \text{ s}</math><br/> If identifies <math>T_A</math> as <math>T_B</math> then 0/2</p> | C1   | [2] |